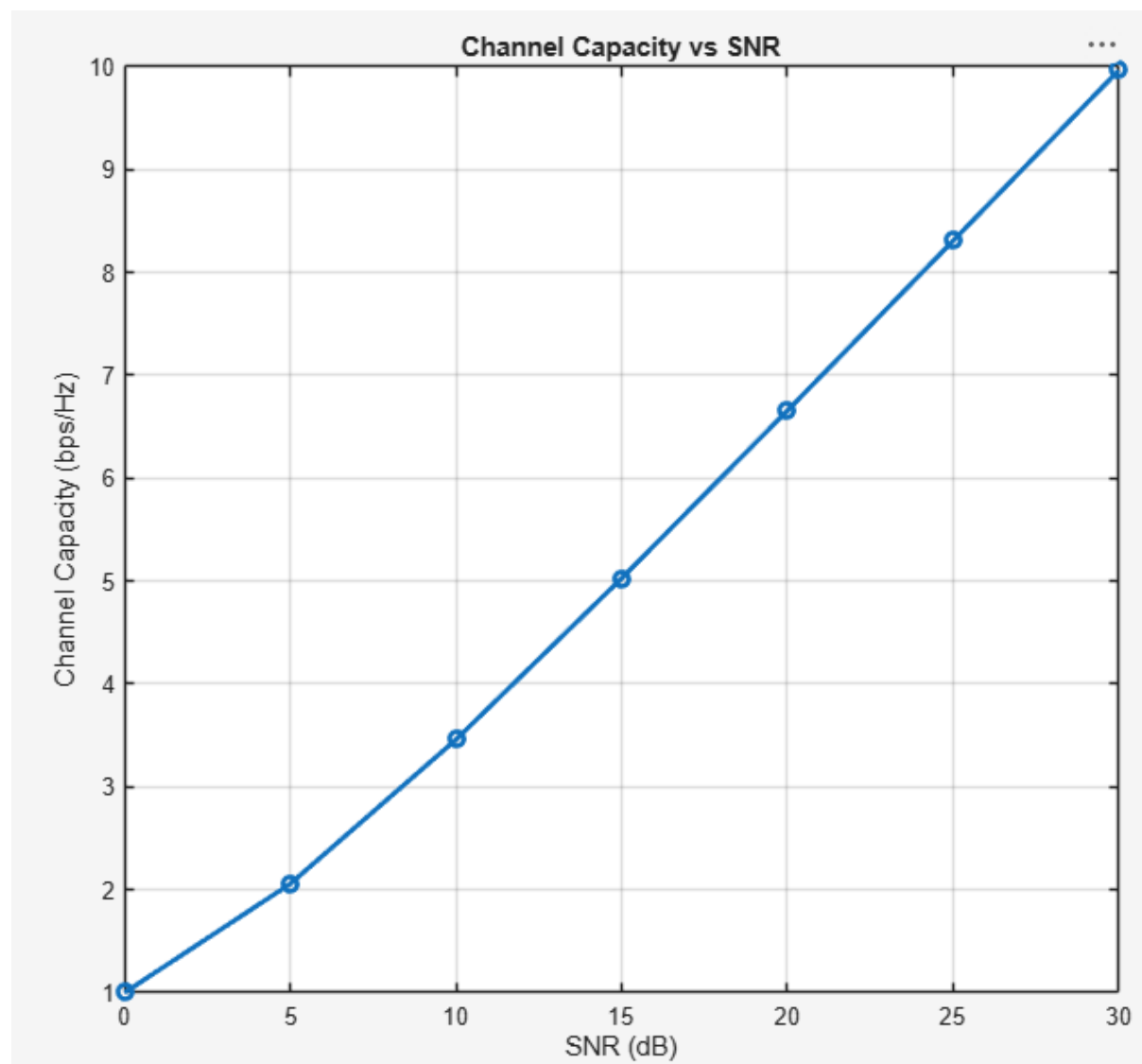


Ex.No.9 Massive MIMO Capacity vs SNR and Gain Using MATLAB

1: Matlab Code and Output

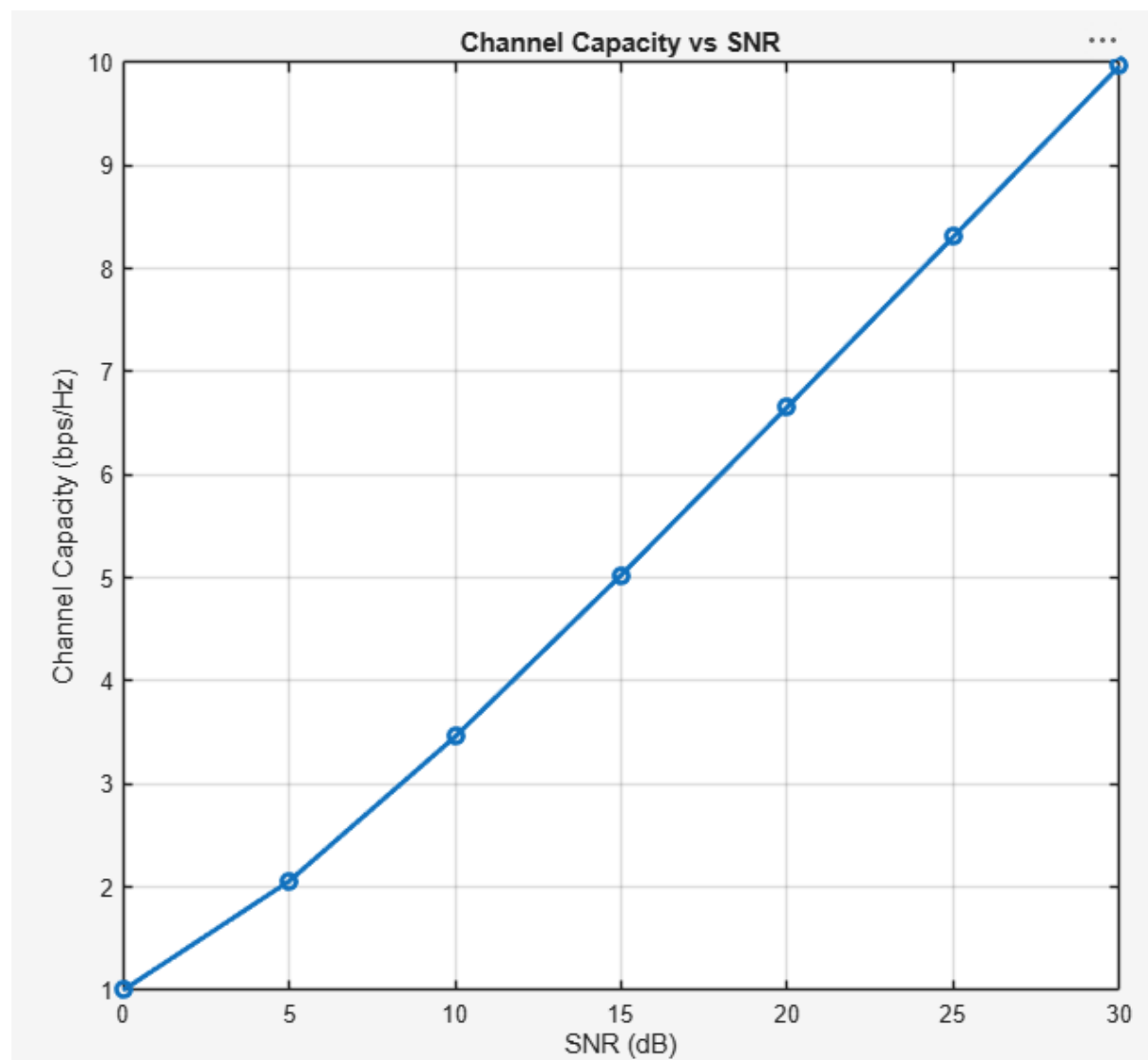
/MATLAB Drive/slotb9.m

```
1  clc;
2  clear;
3  close all;
4  snr = 0:5:30;
5  snr_linear = 10.^(snr/10);
6  capacity = log2(1 + snr_linear);
7  figure;
8  plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
9  grid on;
10 xlabel('SNR (dB)');
11 ylabel('Channel Capacity (bps/Hz)');
12 title('Channel Capacity vs SNR');
13
```



Objective 2: Matlab Code and Output

```
14 snr = 0:5:30;  
15 snr_linear = 10.^(snr/10);  
16 capacity = log2(1 + snr_linear);  
17 figure;  
18 plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);  
19 grid on;  
20 xlabel('SNR (dB)');  
21 ylabel('Channel Capacity (bps/Hz)');  
22 title('Channel Capacity vs SNR');  
23
```



Objective 3: Matlab Code and Output

```
24     antennas = [4 8 16 32 64];
25     gain = 10*log10(antennas);
26     figure
27     bar(antennas, gain)
28     grid on
29     % Number of Antenna Elements
30     % Antenna Gain (dB)
31     xlabel('Number of Antenna Elements')
32     ylabel('Antenna Gain (dB)')
33     title('Antenna Gain for Different Antenna Array Sizes')
```

